

KIRKSVILLE, MO, USA

WMO: 724455

Lat: 40.097N

Lon: 92.543W

Elev: 966

StdP: 14.19

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 14938

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.4	4.6	-8.6	3.6	1.8	-4.4	4.5	6.3	24.9	33.5	22.8	32.8	7.3	300	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	93.0	75.9	90.2	75.7	87.7	74.3	79.0	88.6	77.4	86.9	75.9	84.4	9.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.1	141.2	85.0	74.6	134.3	83.3	73.0	127.1	81.4	43.1	88.9	41.5	86.3	40.0	84.6	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.9	19.8	17.9	-8.9	96.8	5.9	3.8	-13.1	99.6	-16.5	101.8	-19.8	104.0	-24.1	106.8		
			-9.4	81.3	5.7	1.3	-13.5	82.2	-16.8	83.0	-20.0	83.7	-24.1	84.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.4	26.4	29.7	41.5	53.0	62.5	71.6	75.7	73.9	66.5	53.9	42.1	30.6
	DBStd	19.60	11.86	12.21	10.99	9.66	8.25	6.09	5.75	5.48	7.83	9.61	10.33	11.32
	HDD50	2654	733	574	310	77	8	0	0	0	1	69	278	604
	HDD65	5643	1196	987	729	372	148	15	2	4	74	362	689	1065
	CDD50	3531	2	7	47	167	395	648	795	740	496	190	40	4
	CDD65	1047	0	0	1	12	70	213	332	279	120	19	1	0
	CDH74	9723	1	0	10	144	567	1890	3338	2519	1095	157	2	0
	CDH80	3516	0	0	0	18	119	633	1376	969	368	33	0	0
	WSAvg	8.5	9.5	9.6	10.1	10.3	8.6	7.6	6.5	6.3	7.0	8.3	9.4	9.2
	PrecAvg	41.2	1.2	1.9	2.6	4.2	5.2	5.7	4.9	4.0	4.2	3.0	2.6	1.7
Precipitation	PrecMax	65.7	2.9	4.3	6.1	7.3	11.5	12.1	15.4	8.3	12.4	8.9	8.7	4.1
	PrecMin	25.4	0.1	0.4	0.5	1.6	1.3	0.9	0.8	0.2	0.6	0.4	0.2	0.1
	PrecStd	10.4	0.7	0.9	1.4	1.4	2.7	3.0	4.2	2.0	2.8	2.0	1.9	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	67.2	75.1	82.3	88.1	93.1	98.7	96.5	91.6	84.2	72.6	63.4
		MCWB	51.1	54.2	61.6	65.9	72.8	76.4	74.9	75.1	74.1	69.3	59.5	57.7
	2%	DB	53.7	59.5	70.0	78.4	83.1	89.5	93.7	92.0	88.2	78.3	67.6	57.3
		MCWB	45.4	50.6	58.0	63.0	69.7	75.3	76.7	75.1	72.7	65.1	57.5	51.3
	5%	DB	48.0	53.2	64.6	74.2	80.2	86.7	90.6	88.9	84.4	73.7	63.1	51.2
		MCWB	42.1	47.0	55.4	60.9	68.1	74.0	77.0	75.0	71.2	63.8	54.7	46.1
	10%	DB	42.6	47.2	59.6	70.1	77.0	83.6	87.7	85.6	80.7	69.9	58.1	46.3
		MCWB	37.9	41.4	51.2	59.0	65.7	72.2	75.1	73.4	68.6	61.3	50.8	41.8
	0.4%	WB	53.7	58.1	64.1	68.3	75.0	79.2	81.1	80.9	76.8	72.2	62.9	59.3
		MCDB	58.0	64.5	72.7	79.1	84.2	88.9	90.1	90.1	87.7	80.7	68.7	63.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.7	51.8	60.3	65.5	72.2	77.1	79.5	78.3	74.8	68.9	59.3	52.7
		MCDB	52.0	57.0	67.0	75.5	80.8	86.5	89.4	87.5	84.9	74.8	66.0	56.2
	5%	WB	41.9	46.7	57.3	63.0	70.0	75.3	78.0	76.7	73.0	65.7	55.8	46.4
		MCDB	47.8	52.9	63.3	71.8	77.8	83.8	87.6	85.4	81.2	71.3	61.5	49.9
	10%	WB	37.9	41.4	51.9	60.2	67.9	73.6	76.4	74.9	70.9	62.5	51.8	42.2
		MCDB	42.3	47.0	58.5	68.2	74.8	82.0	85.0	82.8	77.9	68.7	57.1	46.4
Mean Daily Temperature Range	5% DB	MDBR	17.0	18.1	19.9	21.2	19.9	19.4	19.9	20.3	22.1	20.6	19.2	16.3
		MCDBR	24.1	25.8	26.4	26.5	22.8	21.9	22.2	23.9	24.4	24.4	24.5	22.3
		MCWBR	17.7	18.5	16.2	15.5	12.1	10.7	9.2	10.3	11.1	13.7	16.3	17.3
	5% WB	MCDBR	22.4	23.7	23.2	22.7	20.0	19.6	19.3	19.7	20.5	19.6	22.3	20.9
		MCWBR	17.4	18.7	15.5	15.0	11.6	10.7	9.7	10.3	10.3	13.2	16.7	17.9
Clear-Sky Solar Irradiance	taub		0.291	0.311	0.333	0.391	0.422	0.423	0.438	0.434	0.387	0.344	0.298	0.292
	taud		2.454	2.382	2.409	2.256	2.233	2.269	2.225	2.258	2.336	2.457	2.524	2.471
	Ebn at Noon		279	287	292	280	272	272	266	264	270	272	275	269
	Edh at Noon		25	31	34	42	44	43	44	42	36	29	23	23
All-Sky Solar Radiation	RadAvg		619	901	1184	1532	1778	1990	1983	1740	1505	1023	698	538
	RadStd		67	78	90	112	135	187	132	94	98	114	62	58

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page